METRIC

A-A-52567
August 5, 1997

COMMERCIAL ITEM DESCRIPTION

SURVEY INSTRUMENT AUTOMATED, INTEGRATED

(AISI)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This specification covers portable, medium-range, 2 kilometer (km) through 7 km, electro-optical distance and angle measuring equipment. Distances are measured with an Electronic Distance Meter (EDM) using diode-emitted, modulated, infrared light waves. Angles are determined by electronic measurement. The AISI measuring unit will be of solid-state construction and complete with accessories for unsupported operation.

2. CLASSIFICATION. The AISI shall be of the following types, as specified:

2.1 Type.

TYPE I - 0°0'0.1"AISI with EDM of 7 km capability.

TYPE II - 0°0'0.1"AISI with EDM of 2 km capability.

2.2 Class.

Class A - With winterization kit.

Class B - Without winterization kit.

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any other data which may improve this document should be sent to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CETEC-PD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

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3. SALIENT CHARACTERISTICS.

3.1 Description. The automated integrated survey instrument, hereinafter referred to as AISI, shall combine into a single, compact field survey total station system the three major subassemblies (modular or integrated) listed below:

a. The Electronic Theodolite (ELT), an auto-reading/recording digital theodolite with electronic display/control panel;

b. The Electronic Distance Meter (EDM), a low power, miniaturized distance measuring device with a range and accuracy compatible with the ELT;

c. The Data Recorder Unit (DRU), an automatic data recording/storage/processing terminal, ruggedized for field use with the ELT and EDM, and interoperable with any microprocessor through an RS232 interface or use industry standard data storage media and provide high-speed data transfer.

The AISI shall be mountable for operation on the standard tripod specified in A-A-52298 using the universal tribrach specified in 3.6.3.1.8 or equivalent. The AISI shall also include transport carry case or cases; retro-directive prisms; two rechargeable battery packs; a battery charger; auxiliary external power cable, not less than 6 meters in length; tribrachs; necessary accessories and tools, with cases and survey system software programs (see 7.2) for computation, plotting and printing. When specified (see 7.2), a winterization kit, thermometer, altimeter-barometer, meteorological correction data, tripods, and a backpack shall also be included. The AISI shall electronically measure values for subtended horizontal and vertical angles; slope distances to a maximum range of 7 km or 2 km, for the type I or type II instrument, respectively; and automatically record these data in a data storage/processing module for on-site reduction or transfer via a serial interface or industry standard interface to another microprocessor. The AISI shall be capable of full daytime and nightime operations. Additionally, the AISI shall be able to cue the layout of horizontal and vertical earthmoving operations when provided with appropriate input data. The AISI should minimize the use of cabling and allow for new features and technologies as they become available.

3.2 Reliability. The mean cycles (see 7.4.1) between failures shall be 1000 cycles, at an 80% confidence level.

3.3 Safety. AISI shall present no uncontrolled hazards to operators, bystanders, or maintenance personnel. The AISI shall meet either the power and irradiance requirements for a class I device listed in TB MED 524, or be less than the irradiance criteria for a class I device and less than five times the power limitations.

3.4 Human factors engineering. The AISI shall conform to human engineering design criteria as described in ASTM F 1166; Section 5 Control/Display Integration, Section 6 Visual Displays, Section 21 Controls, General Information, Section 28 Label Content, Section 33 Maintainability, Section 47 Hazards and Safety, and Section 48 User-Computer Interface.

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3.5 Maintainability.

3.5.1 Maintenance support. Assemblages or support elements such as repair parts, special tools, and test equipment shall be adequate to perform the intended functions of assisting in or conducting maintenance operations.

3.5.2 Maintenance Ratio. The AISI shall have a maintenance ratio (see 7.4.2) of not more than 0.012.

3.5.3 Ease of maintenance. AISI construction shall provide ease of access to batteries and fuses (or reset buttons) such that tools are not required to perform maintenance. AISI construction shall provide a self-test system with a display to verify satisfactory operating capability prior to initial measurement.

3.6 Physical characteristics.

3.6.1 Portability. AISI shall be furnished in soldier portable packages.

3.6.2 Weight. The basic equipment weight (ELT, EDM, DRU, tribrach and rechargeable battery) shall not be more than 11.35 kg. The transport weight (basic equipment and transport carry case) shall not be more than 18.18 kg.

3.6.3 Electronic theodolite (ELT). The ELT, the electronic horizontal and vertical angle measuring component of the AISI, shall automatically display the observed angle at a precision and repeatability required to satisfy 2d order class II accuracy of NOAA Publication (Standards and Specifications for Geodetic Control Networks, September 1984).

3.6.3.1 Alidade assembly. The alidade assembly shall include a self-leveling device, and a means for attaching to the tribrach.

3.6.3.1.1 Telescope. The telescope shall have an aiming device, sunshade, and a focusing range of 4.6 meters to infinity. The telescope shall produce an erect image.

3.6.3.1.1.1 Reticle. The reticle shall consist of a pattern of opaque lines similar to Figure 4. The stadia ratio shall be 1/100. The horizontal and vertical collimation adjustment of the reticle shall be positive and easily accessible. The horizontal and vertical collimation errors of the lines of sight shall be not greater than 0°0'6".

3.6.3.1.1.2 Telescope eyepiece. The eyepiece shall be equipped with a diopter scale marked with plus and minus units from zero, and shall have an index mark for referencing the setting.

3.6.3.1.2 Standards. The horizontal axis of the telescope shall be so nearly perpendicular to the vertical axis of the AISI that the error due to the difference in height of the standards shall not be greater than 0°0'12" or shall be compensated for by onboard software.

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3.6.3.1.3 Circles.

3.6.3.1.3.1 Horizontal and vertical circle. The horizontal and vertical circles shall provide a means of automated correction for circle eccentricity. The resolution of the circle readings shall be 0°0'0.5". The time until initial display shall not exceed one second. Inclination of the vertical axis shall be corrected automatically. Vertical angle measurements shall be automatically indexed within a sensing range of ± 0°10'0" to an accuracy of ± 0°0'0.5". The units of angular measurement shall be operator selectable in either sexagesimal or decimal.

3.6.3.1.3.2 Setting circles. A horizontal and vertical circle setting capability shall be furnished for setting the circles in various positions of an observation program. The circles shall be provided with a means of selecting the units of angular measure so that they correspond to those selected on the microprocessor.

3.6.3.1.4 Microprocessor display/control panel.

3.6.3.1.4.1 Keyboard. Panel dials, displays, knobs and switches shall be functionally grouped for simple operation. Each shall be marked to identify its functional name, symbol and operating instruction reference. The keyboard shall be weatherproof and shall be capable of initiating the microprocessor functions listed in 3.6.3.1.4.4.

3.6.3.1.4.2 Displays. The display shall depict operational commands, data entry commands, self-test status, and data input, as a minimum. Displays shall be legible in any ambient light.

3.6.3.1.4.3 Illumination. The electronic control panel shall have internally mounted illumination for night operation. All external lighting shall be a blue-green color between 400-700 nanometers in wave length. No light shall be visible beyond a 50 meter circle with the instrument as center point, under moonless and starless night conditions. All externally visible light produced shall be intensity adjustable.

3.6.3.1.4.4 Microprocessor. The AISI microprocessor shall have a nonvolatile memory of adequate capacity to store all instrument constants, operating parameters and enough random access program memory to perform required data reductions. Upon start-up, the microprocessor shall initiate and control a system self-test, that shall include power source check, EDM operational check, and display test sequence. The microprocessor shall be capable also of executing the following listed functions from the control panel keyboard input:

OPERATIONAL COMMANDS
Power on-off
Illumination on-off
System self-test, if not automatically executed.
Clear entries/commands
Select/display units of angular/distance measure
Select precision of angular display readings
Set horizontal circle orientation
Measure/display/record horizontal angle or direction
Measure/display/record vertical or zenith angle

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Measure/display/record horizontal collimation error
Measure/display/record inclination of standing axis
Level standing axis
Measure/display/record slope or horizontal distance
Measure/display/record height differences or elevations
Display coordinates
Display stake-out direction/distance
Display error/system messages

DATA ENTRY COMMANDS
Enter vertical circle index error
Enter horizontal collimation error
Enter stake-out data
Enter initializing horizontal/vertical datum
Enter height of instrument/target
Enter instrument/target constants
Enter scale corrections
Enter atmospheric data/corrections
Enter system codes (stationing, record)
Enter system format parameters (bits per second (BPS), parity etc.) for data transmissions
Enter instrument/target eccentricity.

3.6.3.1.5 Axes motion controls. Movement through the vertical tilt limits shall be a minimum of ±45°. Horizontal and vertical motion controls shall provide for fine and coarse adjustment.

3.6.3.1.6 Optical plummet. The optical plummet shall be integral with the alidade or shall be attached to the tribrach and shall be adjustable. The reticle pattern shall divide the field of vision into four 90°0'0" quadrants. The optical system shall produce a clear image throughout the operating range from tripod height to not less than 30.5 meters. The line of sight of the optical plummet shall be coincident with the vertical axis of the AISI. A means of illuminating the reticle and controlling the light intensity shall be provided.

3.6.3.1.7 Levels. The instrument shall be equipped with leveling capability that will allow the instrument to be setup and operated at the required accuracies.

3.6.3.1.8 Tribrach. The tribrach shall consist of the leveling screws, base plate, and a device for locking the alidade assembly. The base plate shall have at its center a 5/8-inch nut of the unified coarse thread series class 2B, 11 threads per inch (5/8-11 UNC-2B) for compatibility with existing equipment. The leveling screws shall permit leveling of the theodolite when the base plate is 6°0'0" out of the level. The design of the tribrach shall be such that removing and returning the alidade shall not cause a change in the centering of the alidade greater than 0.05 mm. If optical plummet is integral to tribrach see 3.6.3.1.6.

3.6.3.2 Power. The ELT/EDM shall operate from an internal or external battery. AISI shall contain built-in protection against reverse polarity from the power source, including a battery compartment that precludes improper battery insertion. AISI winterization kit heating elements may operate from an external battery.

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3.6.3.2.1 System power. The ELT/EDM shall be provided with two battery packs. Each battery shall be capable of supporting a minimum of 500 angle and distance measurements at 20 ºC. The batteries shall be rechargeable (see 3.6.3.2.3), spillproof and shall have no detectable liquid or gas seepage regardless of operation or storage orientation.

3.6.3.2.2 Auxiliary power. The ELT/EDM shall also operate from external 12-volt DC batteries, providing the same operating capability as with the internal power source. An external power cable shall be provided with the system that shall not be less than 6 m in length and shall terminate in automotive type battery clamps which are identified as positive and negative.

3.6.3.2.3 Battery charger. System batteries shall be rechargeable using a 110/220 volt, 50/60 Hz charger that shall be supplied with the system. The battery charger shall be capable of concurrently charging at least two battery packs or a built-in battery charger shall be furnished with each battery pack. The battery shall not be damaged nor lose capacity when charged continuously from the charger supplied with the system.

3.6.4 Electronic distance meter (EDM). Distance measurements shall be acquired in either a normal mode (individual measurements manually cued and displayed), tacking mode, (continuously measured and displayed), or distance averaging mode. The EDM shall be capable of operating from the power supplied through the ELT, or using power directly from a separate external power source (see 3.6.3.2.2). The keyboard control shall allow for the entry of the following commands: Power/illumination on-off; measure and display distance; activate tracking mode; stop tracking mode; self-test (signal strength, power level) and data entry (prism constant, scale correction). The units of distance measurement and data entry shall be keyboard selectable in either the Metric or English system. The keyboard shall provide the above functions and in addition, extend the capabilities of the ELT/EDM combination by providing for remote EDM control, horizontal distance display, height difference, differential ground point stake-out, coordinate differences, and atmospheric corrections.

3.6.5 Data recording unit (DRU). This component of the AISI shall automatically acquire data from the ELT and down-load through an appropriate interface to another data processing unit. Additionally, it shall be operator programmable to perform data storage, transfer and customized computing tasks. The programming language shall be user friendly. Prepared programs shall be supplied to accomplish common field routines such as coordinate geometry calculations, stakeout, precise leveling and construction alignment. A means of reference to frequently entered field data codes, DRU commands, or program instructions shall be attachable to the DRU for operator convenience. Non-volatile memory capacity shall be adequate to store data transfer parameters (bits per second (BPS), parity, etc.) and field station data (horizontal and vertical angle, horizontal distance, and station designation/information code) for a minimum of 500 survey stations. The power source necessary to maintain memory non-volatility and shall have a minimum supporting life of 12 months.

3.6.5.1 Data structure. The DRU shall be capable of recording, accessing, editing and executing program files. The data organization shall allow separate storage and retrieval of data files containing station identification information, horizontal angle, horizontal distance and vertical angle parameters for a minimum of 500 ground control points. The design, size and number of both data and program files shall be sufficient to support this field survey data collection,

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encoding and program commands. The DRU shall be capable of data transfer using interfaces such as RS232 or PC card. Transfer rate shall be a selectable transmission at a minimum of 110 to 9600 BPS. Additional data transfer information such as parity, character format, protocol and terminal emulation shall be selectable.

3.6.5.2 Display. Displays shall be legible in any ambient light as well as indirect or artificial light. Multiple or single displays shall be capable of the presentation of all operator prompts, program instructions and data.

3.6.5.3 Power. The DRU shall be powered by a rechargeable power supply. Operating power loss shall not affect any data stored in the non-volatile memory and shall be preceded by a display warning.

3.6.6 Reflector set. Two retro-reflector prism sets shall be provided as a component of the AISI. Each set shall consist of the following.

a. Sufficient number of prisms to satisfy maximum range capabilities for type I and type II AISI's (see 3.11.3.2).
b. Single and multi-prism holders to accommodate all prisms provided.
c. Prism adapter to allow mounting of a combination of prism holders.
d. Mounting adapter for mounting both the prism holders and adapter in the universal tribrach specified in 3.6.3.1.8. The mounting adapter shall permit the prisms to be rotated through 360 degrees of arc about the vertical axis without disturbing either the tribrach or plumb of the assembly over the survey mark. A clamping device shall be provided for locking the prisms at any point after rotation.
e. Transport carry case or cases conforming to 3.6.7.

3.6.7 Transport carry cases. Each transport carry case (housing the electro-optical package, accessories and tools) shall be of a rigid sturdy construction with a removable inner liner, molded to accept the various AISI accessories. The cases shall withstand operating, transport, and storage usage and shall be adequately blocked and padded to prevent damage to the contents. The cases shall prevent any damage to the AISI when dropped from a height of 91 cm. The cases shall also protect the instrument against damage from rainfall, shock, and vibration as specified herein. Each case shall contain a desiccant. Each case shall open in such a way that the contents will not fall out when the case is opened. The cases shall be provided with heavy-duty locking catches or means for padlocking. All items listed in 3.6.9 (except the backpack and tripod) shall be stored in transport carry cases. The cases shall be the same color as the instrument and shall be no larger than 63.5 cm x 30.5 cm x 25.4 cm. Each transport carry case shall weigh no more than 11.35 kg and shall be labeled with a loading plan (see 3.7)

3.6.8 Accessories and tools. The contractor may provide additional items as required to perform operator maintenance and field adjustments. Accessories and tools shall be provided with a separate case similar in construction to type indicated in 3.6.7, or packed with major components in their respective transport carry cases (the pole, and plumbing may be packed in its own case). The following shall be the minimum furnished with each AISI as accessories and tools:

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Item	Quantity
Spare Fuses (when used)	4 each
Brush, lens	1 each
Tissues, lens	1 set
Operators manual	1 each
Covers, lens, ELT, (both ends)	1 set
Cover, lens, EDM	1 set
Cover, instrument, waterproof	1 each
Pole, prism; 2.44 meters (adjustable), with level	2 each

3.6.9 Optional equipment. The following equipment will be provided with the AISI when specified in ordering data (see 7.2):

Item	Quantity
Backpack (see 3.6.9.1).	2
Winterization kit (see 3.6.9.2). including:	1
Thermometer, centigrade (commercial std)	1
Altimeter/barometer, (mm Hg graduation) commercial std, with meteorological correction charts.	2
Tripod (see 3.6.9.3).	3

3.6.9.1 Backpack. When specified (see 7.2), a means for transporting AISI by backpacking shall be provided. The backpack shall be water-repellent and a subdued color or camouflaged. The backpack shall be easy to use and shall be comfortable to wear for male and female personnel ranging anthropometically from the 5th to 95th percentile.

3.6.9.2 Winterization kit. When specified (see 7.2), a winterization kit shall be furnished for use with AISI at temperatures below -20 ºC. The winterization kit shall include covers for all control knobs and leveling screws (unless plastic or plastic coated), shields for the telescope, and the optical plummet eyepieces. The kit shall be contained in a pliant, water-repellent carrying case. The kit shall be capable of heating AISI to required operating temperature from -32 ºC within 25 minutes and from -43 ºC within 45 minutes. An auxiliary power source (see 3.6.3.2.2) may be used.

3.6.9.3 Tripod. When specified (see 7.2), tripods conforming to A-A-52298 shall be furnished.

3.7 Identification marking. A loading plan with the identification of stored components shall be permanently attached to the top inside cover of the transport carry case, or the inserts of the transport case shall include cut-outs that indicate where the item should be placed.

3.8 Interchangeability. All components from the same manufacturer having the same system part number or function shall be functionally and dimensionally interchangeable.

3.9 Treatment and painting. Untreated, exposed, exterior surfaces of the AISI, cases, and accessories shall be painted in accordance with the manufacturers best commercial practices. Unless otherwise specified, (see 7.2), the color shall be lustreless army green.

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3.10 Lubrication. All portions of the AISI requiring lubrication shall be lubricated prior to assembly. The lubrication shall permit the AISI to operate throughout the specified temperature ranges.

3.11 Performance.

3.11.1 Calibration. Each AISI shall be calibrated so that there are no instrument constant or retrodirective prism offsets or effects that cannot be corrected prior to distance and angle measurement. If such constants, offsets, or effects do exist, they shall be recorded on the applicable hardware in a conspicuous place to facilitate presetting AISI. A means shall be provided for adjusting AISI to the proper constant without use of special tools.

3.11.2 Electronic theodolite.

3.11.2.1 Measurement accuracy. The measurement of horizontal and vertical angles shall yield a Standard Deviation of mean pointings between direct and reverse sightings of 0°0'0.8" or less. Maximum resolution of displayed angular values shall be 0°0'0.1".

3.11.2.2 ELT and EDM measurement time. After setup and actuation of the AISI's off-on power switch to the ON position, the time required for an initial distance and angular reading in ambient temperatures from 40 to -20 ºC shall be not greater than 4 minutes. Subsequent distance and angular readings shall be accomplished in not greater than 6 seconds. The same time limitation shall apply to AISI after having been warmed from -43 to -20 ºC by the winterization kit as specified in 3.6.9.2, when applicable.

3.11.2.3 Data processing, storage, transfer. Field measurement data shall be automatically and manually manipulated in accordance with 3.6.3.1.4.4 without loss of information or identification.

3.11.3 Electronic distance meter.

3.11.3.1 Measurement accuracy. The maximum allowable mean-square error for a mission of 10 observations, at an ambient temperature range of -5 to +50 ºC, shall not exceed ±5mm, +5ppm, in standard mode and ±10 mm, +5ppm in tracking mode. (Note: Temperatures shall be accurate to ±1 ºC and pressures to ±5mm of mercury.)

3.11.3.2 Range. Atmospheric conditions and the number of prisms used as targets determine the range capabilities of the EDM. Minimum range capability under all atmospheric conditions shall not exceed 2 meters. The following maximum ranges shall be achieved at the accuracy specified in 3.11.3.1 under the conditions stated for type I and type II AISIs.

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Table I. Atmospheric conditions.

		Poor	Average	Excellent
Type I	1 Prism 3 Prism 7 Prism 11 Prism	1.0km 1.2km 1.7km 1.8km	2.0km 3.5km 4.5km 5.0km	2.5km 5.0km 6.0km 7.0km
Type II	1-3 Prism	0.5km	1.5km	2.5km

Atmospheric conditions are defined as:

Poor - Visibility 3km with strong haze, bright sun, and heat radiation.

Average - Visibility 15-20km, light haze, moderate sun, light heat radiation.

Excellent - Visibility 40km, clear, overcast to partly cloudy, no heat radiation.

3.12 AISI system software. The system software shall be a fully integrated, three dimensional, ground modeling, design, and drafting system. The system should include such applications as field note reduction, automated data collection, traverse balancing, surface modeling, contour mapping, coordinate geometry, roadway design, drainage design, earthwork volumes by average-end-area or true prismoidal calculations, and three dimensional projections. The software shall be executable from MS-DOS type system or Windows and shall run on IBM and IBM compatible hardware. The software may contain separate applications to perform diverse functions such as drafting and design calculations. However, maximum integration into one package is highly desirable. Software application shall be in the two areas of topographic (control) survey and construction survey. In both areas the software shall be required to transfer data between recorder and processor and to peripheral equipment such as printer and plotter. Representative tasks to be performed in the topographic area include: storage of raw field data; compute, adjust, plot and annotate three dimensional coordinates using coordinate geometry (includes traverse, intersection, and resection); compute a level line; compute astronomic azimuth (stellar); coordinate conversion universal transverse mercator (UTM) to geographic, world geodetic system (WGS); state plane coordinate conversion; zone to zone conversion (UTM); and compute convergence of coordinate lines. In the construction survey task area the following representative tasks are critical: storage of raw field data; compute, adjust, plot, and annotate three dimensional coordinates using coordinate geometry (includes traverse, intersection and resection); compute and adjust level data; compute and format for field location, road, airfield, buildings, and utilities (centerline, grade, earthwork); compute and plot profiles, cross sections, vertical curves, and horizontal curves; compute and plot preliminary/final road, airfield and architectural construction plans. Software must be interactive, menu driven, and user friendly to the maximum extent possible.

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3.13 Environmental requirements.

3.13.1 Operational temperature. The AISI ,without use of the winterization kit, shall operate with the accuracy specified herein when in thermal equilibrium in air at any temperature between -20 and 50 ºC. The AISI, with the use of the winterization kit if necessary, shall operate with the accuracy specified herein from -42 to -17 ºC.

3.13.2 Storage temperature. The AISI shall not be damaged by storage at temperatures as low as -46 °C for 6 hours and as high as 71°C for 4 hours after stabilization (see 7.4.4).

3.13.3 Solar radiation. The AISI shall operate without damage within the accuracy specified herein when subjected to solar radiation of 355 Btu/ft squared/hour, for not less than 4 hours in an ambient air temperature of 43 +0, -2 ºC.

3.13.4 Humidity. The AISI shall operate within the accuracy specified herein and shall not be damaged when subjected to relative humidities of not less than 95 percent, at air temperatures between -17 ±1, and 50 ±1 ºC.

3.13.5 Rain. The AISI mounted in the operating position, with an instrument cover in place, shall show no evidence of water penetration after exposure to windblown rainfall of 19.35 cm/hr at wind speeds not less than 48 k/h for a period of not less than 60 minutes. An instrument cover is allowable.

3.13.6 Vibration. The AISI, packaged in its storage case, shall endure 45 minutes of vibration in each of the 3 axes with a 2.5-3.0 cm double-amplitude orbital path at 5 Hz.

3.13.7 Shock. The AISI, packaged in its storage case, shall endure one drop of 91 cm on each of the following: each of the 6 faces, each of the 8 corners, and each of the 12 edges, for a total of 26 drops.

3.13.8 Magnetic environment. AISI shall operate and measure distance within the accuracy specified in 3.11.3.1 and time specified in 3.11.2.2 while in a magnetic field of three gauss. The magnetic field orientation shall be east-west, north-south, and vertical.

3.13.9 Blowing dust. The AISI shall endure a blowing dust environment in the storage mode. with the following criteria:

a. The air velocities shall be 1.5 to 8.9 m/s.
b. The dust composition shall be consistent with red china clay or silicon flour.
c. The dust concentration shall be maintained at 10.6 ±7 g/m^3.
d. The AISI shall be oriented so that all surfaces shall face the blowing dust equal amounts of time.
e. The air temperature shall be 23° C for six hours and 50° C for six hours..
f. The AISI shall be configured in the storage mode.

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3.13.10 Blowing sand. The AISI shall endure a blowing sand environment in the storage mode with the following criteria:

a. The air velocities shall be 18 to 29 m/s.

b. The suggested sand composition (at least 95% silicate oxide by weight) is as follows:
   1.00% shall be retained by a 20 mesh screen.
   2.00% shall be retained by a 30 mesh screen.
   15.00% shall be retained by a 40 mesh screen.
   37.00% shall be retained by a 50 mesh screen.
   30.00% shall be retained by a 70 mesh screen.
   15.00% shall be retained by a 100 mesh screen.

c. The sand concentration for natural conditions is 2.2 ±0.50 g/m^3.

d. The case shall be oriented so that all surfaces shall face the blowing sand for 90 minutes per surface.

e. The air temperature shall be 50° C.

3.14 Electromagnetic interference. The instrument shall have design characteristics for the control of electromagnetic interference emission and susceptibility.

3.14.1 Conducted emissions. Conducted emissions (CE) on power leads and returns, that obtain power from sources not part of the unit shall not exceed the values in figure 1 for a frequency scan from 10 kHz to 10 MHz.

3.14.2 Conducted susceptibility. Conducted susceptibility (CS) on all interconnecting cables, including power cables, shall not exceed the values in figure 2 for a frequency spectrum scan from 10 kHz to 10MHz. Use curve A for frequency ranges 10 kHz to 2MHz, and curve B for frequency ranges from 2MHz to 10MHz.

3.14.3 Radiated emissions. Radiated emissions (RE) from the instrument and all interconnecting cables shall not exceed the values in figure 3 for a frequency spectrum scan from 2 MHz to 1 GHz or 10 times the highest intentionally generated frequency within the instrument, whichever is greater. Measurements beyond 18 GHz are not required.

3.14.4 Radiated susceptibility. Radiated susceptibility (RS) shall not exceed 50.00 volts/meter for a frequency spectrum scan from 2 MHz to 18 GHz. The requirements shall be met for vertically polarized fields up to 30 MHz and for both horizontally and vertically polarized fields above 30 MHz. Circular polarized fields are not acceptable.

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4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials (see 7.4.3) to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS. The supplier shall comply with the following for certification and acceptability.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The AISI offered shall be of the latest model of the standard product of the supplier and shall have been used in the field for at least one year by commercial organizations.

5.3 Examination. Each AISI shall be examined for compliance with the requirements specified in 3.2 through 3.14. No lateral motion shall exist between the electro-optics housing and support. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between glass and coated film, or cloudiness. Optics shall show sharpness of focus and distinct images to provide favorable definition. Focusing components or assemblies shall retain their focus when finger pressure is released and the telescope shall not change focus when transited. Metal surfaces on the optical components for all prisms shall not shift. There shall be no flickering of the illumination system when the vertical or horizontal motions are rotated. Liquid within the level vials shall be clear, and the vials shall be tight in the housings. Mirrors shall not be tarnished, and backing shall be uniform and unspotted. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 When this Commercial Item Description is used for procurement the product conformance clause must appear in the solicitation.

7.2 Acquisition requirements.

a. Title, number, and date of this document.
b. Type of AISI required (see 2.1).
c. Class of AISI required (see 2.2).
d. When color other than lustreless army green is required (see 3.9).

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e. When a backpack is required (see 3.6.9.1).
f. When tripods are required (see 3.6.9.3).
g. Number of survey system software packages per AISI required (see 3.6.5)
h. Level of preservation and packing required.

7.3 Source of documents.

ASTM F 1166 is available from the American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103.

A-A-52298, and MIL-STD-461, are available from the Defense Automated Printing Service, DODSSP, 700 Robbins Avenue Bldg 4/D, Philadelphia, PA 19111-5094.

TB MED-524 is available from the Commander, U.S. Army Publication Distribution Center, 1655 Woodson Road, St Louis, MO 63114.

Standards and Specifications for Geodetic Control Networks, September, 1984, is available from the National Geodetic Information Branch, N/OG174 Rockwall Building, Room 24, National Geodetic Survey (NOAA), Rockville, MD 20852.

7.4 Definitions. The following definitions apply to this document.

7.4.1 Cycle. A cycle is defined as the performance of steps a. through d. below.

a. Unpack and setup equipment for operation over certified baseline with targets of known angular value observable.
b. Perform ten distance and six horizontal angle measurements to selected targets within the maximum range limits of the type AISI being tested. Angular measurements shall consist of six pointings in the direct telescope position and six in the reverse telescope position. Measurement accuracies shall conform to the requirements in this document.
c. Dismantle and repack equipment for transport.
d. Perform manual transport of not less than 2 meters and return to original position.

7.4.2 Maintenance ratio. The maintenance ratio is expressed as the cumulative number of man-hours of maintenance expended during a given period divided by the cumulative number of man-hours of operation during the same given period.

7.4.3 Recovered materials. For the purpose of this requirement, recovered materials are those which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. Used, rebuilt, or remanufactured component, pieces and parts shall not be incorporated in the AISI.

7.4.4 Stabilization. Stabilization is a 2° C or less differential between the ambient temperature and the instrument.

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7.5 Subject term (key word) listing.

Theodolite
Electronic distance measurement
Transit

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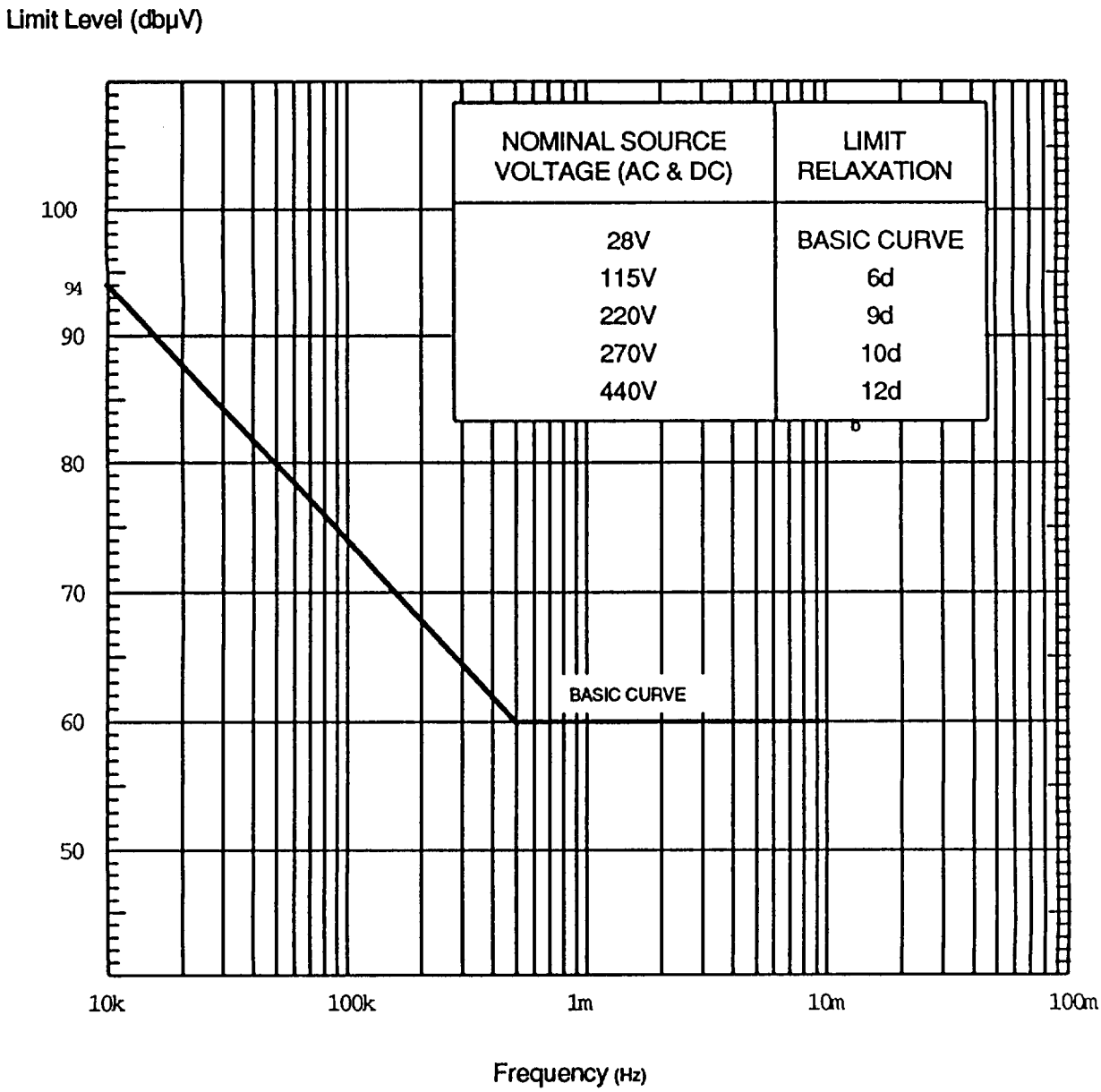


Figure 1. Conducted Emission Limit.

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Check the source to verify that this is the current version before use.

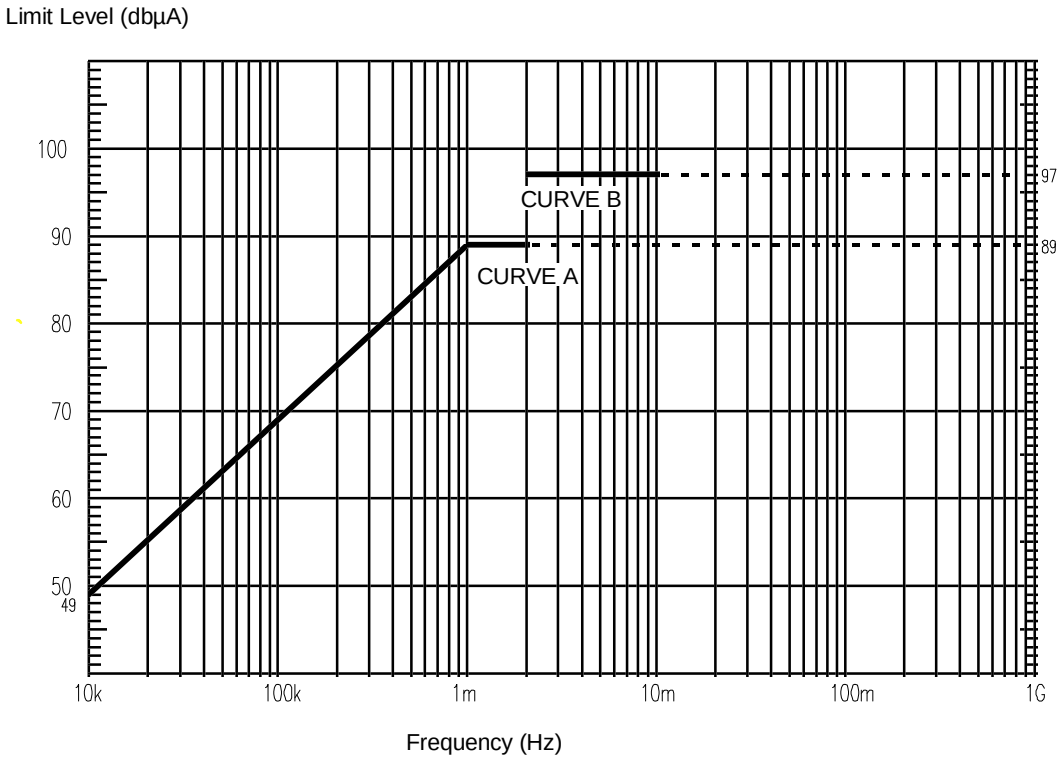


Figure 2. Conducted Susceptibility Limit.

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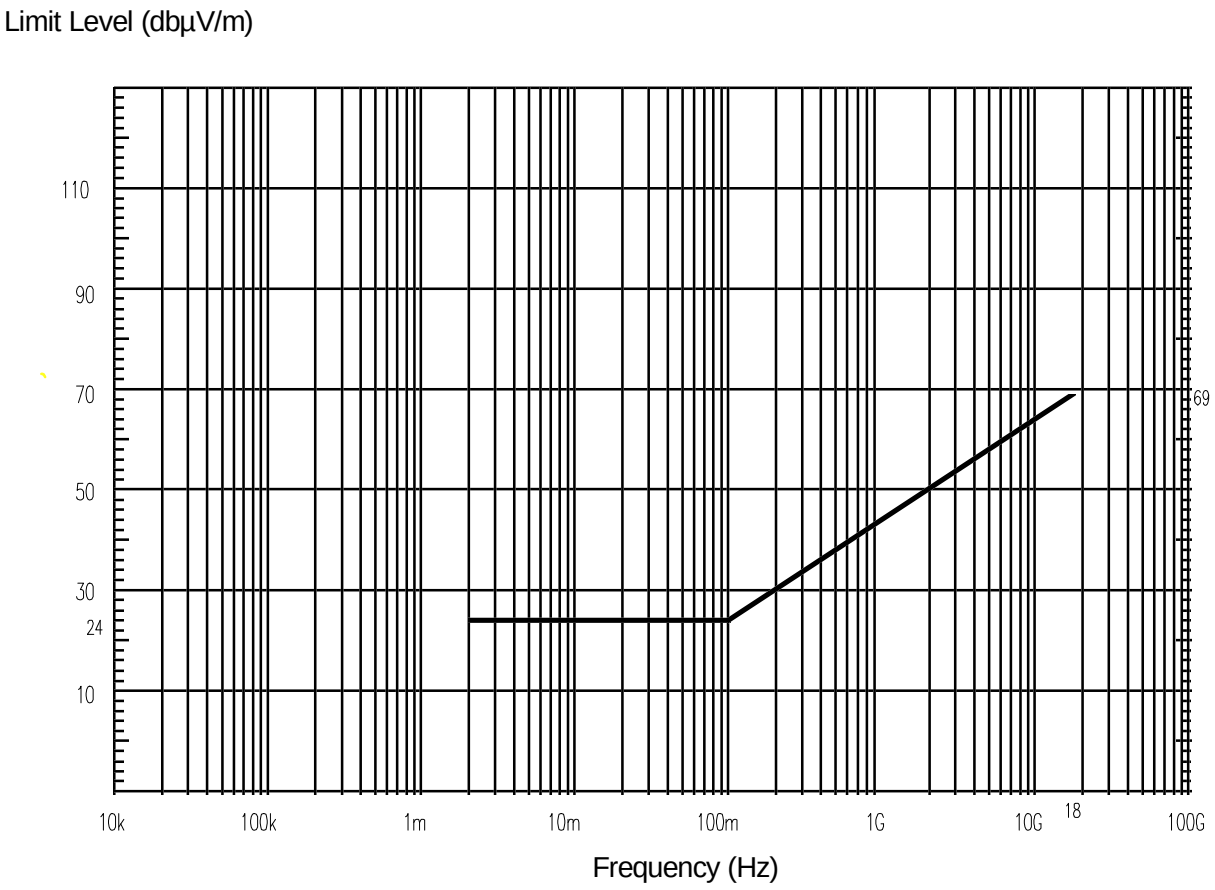


Figure 3. Radiated Emission Limit.

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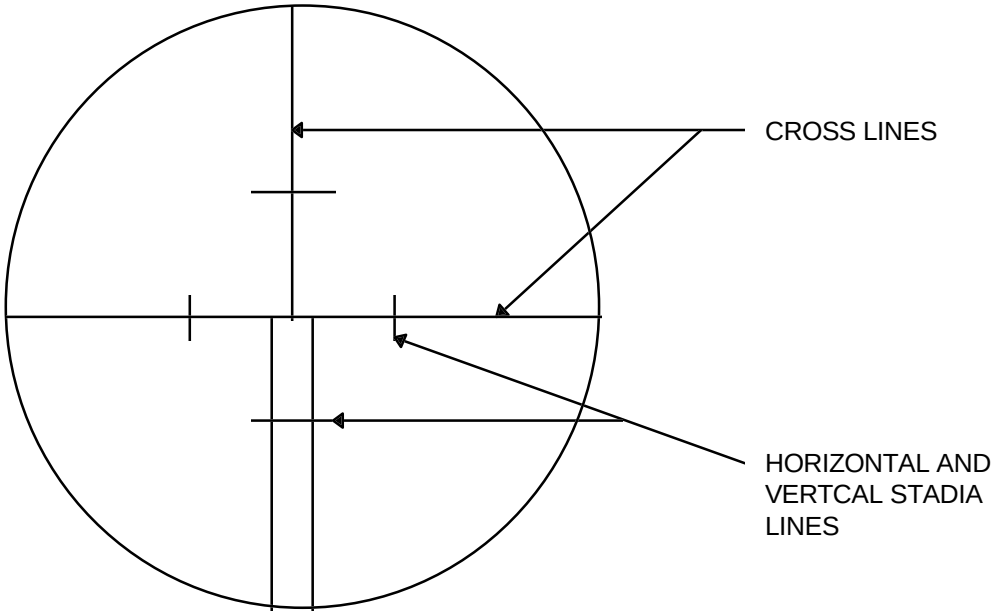


Figure 4. Reticle Pattern for Theodolite.
(Illustrative example only)

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Custodians:
Army - CE3
Navy - YD1

Preparing Activity:
Army - CE3
Project 6675-0047

Review Activities:
DLA - GS

Civilian Coordinating Activity:
GSA - FSS

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